

Overview

Melody Huang

ACIC 2026 Short Course

`melody.huang@yale.edu`

Link: `https://tinyurl.com/acic2026-hte`

Motivation

- Much of causal inference has focused on the **average treatment effect** (ATE).
- However, the ATE may not be informative about the effects of treatment on specific subgroups of the population.
- This motivates the study of **heterogeneous treatment effects** (HTE), which studies how treatment effects vary across different individuals.
- A few examples:
 - **Get-out-the-vote:** Are there different canvassing messages that induce voters to **change their behavior differently**? How do voters respond differently?
 - **Jobs training program:** Who **benefits more/less** when they receive access to a training program? Who should these programs try to recruit in practice?
 - **Clinical trials:** are there people who might be **adversarially harmed** from receiving the treatment? How can we understand potential moderators in a data-driven approach?

What is this short course about?

We focus on how to **interpret** the results of heterogeneous treatment effect estimation, and how to evaluate the performance of different heterogeneous treatment effect models.

- Not about how to construct new heterogeneous treatment effect estimators!

Overview of topics:

- Review of heterogeneous treatment effect estimation
- Interpretability and subgroup discovery
- Diagnostic tools

Notation

- $Y(1), Y(0)$: potential outcomes under treatment and control, respectively
- X : pre-treatment covariates
- Z : treatment assignment indicator
- Estimands:
 - Average Treatment Effect (ATE):

$$\tau = \mathbb{E}[Y(1) - Y(0)]$$

- Conditional Average Treatment Effect (CATE):

$$\tau(X) = \mathbb{E}[Y(1) - Y(0) \mid X]$$

- Identification assumptions:
 - Conditional ignorability of treatment assignment

$$Y(1), Y(0) \perp\!\!\!\perp Z \mid X$$

- Positivity: $\eta \leq \pi(X) \leq 1 - \eta$

Identification of the CATE

Under conditional ignorability and positivity, we can identify the CATE:

$$\begin{aligned}\tau(X) &= \mathbb{E}[Y(1) - Y(0) \mid X] \\ &= \mathbb{E}[Y(1) \mid X] - \mathbb{E}[Y(0) \mid X] && \text{(Linearity in Expectation)} \\ &= \mathbb{E}[Y(1) \mid X, Z = 1] - \mathbb{E}[Y(0) \mid X, Z = 0] && \text{(Cond. Ignorability/Positivity)} \\ &= \mathbb{E}[Y \mid X, Z = 1] - \mathbb{E}[Y \mid X, Z = 0] && \text{(SUTVA)}\end{aligned}$$

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What about **estimation**?

- Basic premise: try to estimate $\tau(X)$ by modeling $\mathbb{E}[Y \mid X, Z = z]$.

T-Learner

- Fit **any** black box machine learning model to construct a flexible model for $\mathbb{E}[Y \mid Z, X]$

$$\hat{m}_z(X) = \mathbb{E}[Y \mid Z = z, X] \text{ for } z \in \{0, 1\}$$

- The CATE can then be estimated as:

$$\hat{\tau}(X) = \hat{m}_1(X) - \hat{m}_0(X)$$

- This class of approaches is often referred to as the **T-learner**.
- Popular approach: BART
 - Use BART to estimate $\hat{m}_1(X)$ and $\hat{m}_0(X)$ and then take the difference between the two models for an estimate of $\hat{\tau}(X)$.

Limitations of the T -Learner

A problem with T -learners is that while a flexible model for $\mathbb{E}[Y \mid Z, X]$ may be good at predicting the **observed** data, it may fail to estimate the CATE very well.

Consider the following:

$$Y(z) = X^\top \beta_z + \epsilon_z$$

Assume researchers use two separate Lasso models to estimate $\{\hat{\beta}_1, \hat{\beta}_0\}$.

- $\hat{\beta}_1$ and $\hat{\beta}_0$ will both be regularized away from zero
- Naively treating CATE estimate as $X^\top (\hat{\beta}_1 - \hat{\beta}_0)$ is problematic
 - Possible to **falsely find treatment effect heterogeneity** even in constant τ settings
 - Also possible to **fail** to pick up on treatment effect heterogeneity
- See Kunzel et al. 2019 here for other examples

Causal Tree

In a standard regression tree setting, a tree is estimated by recursively partitioning X by finding the covariate/split point that minimizes the mean-squared error of the prediction target.

→ **Problem:** in HTE settings, we cannot compute the MSE of τ_i !

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→ **Problem:** in HTE settings, we cannot compute the MSE of τ_i !

Instead, Athey and Imbens (2016) use a modified splitting criterion, the **Expanded Mean Squared Error (EMSE)**:

$$\text{EMSE} = \mathbb{E} \left[(\tilde{Y} - \hat{\tau}(X))^2 - \tilde{Y}^2 \right],$$

where $\tilde{Y} = Y/(Z\pi(X) - (1 - Z)(1 - \pi(X)))$ is a auxiliary outcome variable.

We can re-write the EMSE:

$$\mathbb{E} \left[(\tilde{Y} - \hat{\tau}(X))^2 - \tilde{Y}^2 \right] = \mathbb{E}[\text{var}(\hat{\tau}(X))^2] - \mathbb{E}[\tau(X)^2]$$

- Minimizes within leaf variance, and maximizes across leaf variance.

X-Learner

Kunzel et al. (2019) propose the X -learner, which constructs a pseudo-effects using the nuisance functions:

$$\tilde{\tau}_i^z = Y_i - \hat{\mu}_z(X_i), \text{ for } z \in \{0, 1\}.$$

- With $\tilde{\tau}_i^z$, estimate a non-parametric regression to construct estimated CATE for both the treatment and control units (notated as $\hat{\tau}_{(1)}(X_i)$, $\hat{\tau}_{(0)}(X_i)$, respectively)
- Estimates are then aggregated together for the final CATE estimator:

$$\hat{\tau}(x) = \{1 - \hat{\pi}(x)\}\hat{\tau}_{(1)}(x) + \hat{\pi}(x)\hat{\tau}_{(0)}(x).$$

R-Learner

The *R*-learner uses the following objective function:

$$g_{\tau}(\cdot) = \operatorname{argmin}_{\tau} \left\{ \mathbb{E}_n \left[(Y - m(X) - \{Z - \pi(X)\} \tau(X))^2 \right] + \Lambda_n(\tau(\cdot)) \right\},$$

where $\Lambda_n(\tau(\cdot))$ is a regularizer on the complexity of the $\tau(\cdot)$ function (Nie and Wager, 2021).

- $m(X)$ and $\pi(X)$ correspond to the oracle functions.
- In practice, researchers employ estimates of $m(X)$ and $\pi(X)$, and a two-step procedure is then used to solve the objective function.

Summary

In general, estimating heterogeneous treatment effects follows the same general procedure:

- Estimate nuisance parameters: $m_1(X), m_0(X), \pi(X)$
- Aggregate them together to construct an estimate of the CATE estimator
 - Different approaches differ in how they estimate the nuisance parameters, and how they aggregate them together.

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In general, estimating heterogeneous treatment effects follows the same general procedure:

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Across all of these approaches, there are **common challenges**.

- Unlike standard prediction problems, the true CATE is never observed, which makes it difficult to evaluate the performance of different methods.
- As X increases in dimensionality, we lose interpretability of the CATE estimates
- Performance of different methods can be highly sensitive to the choice of tuning parameters and the quality of the nuisance function estimates!